

Q: Darling Mills Creek leads to which river, and what does that river flow into?

Evidence-Link Construction



Endpoint-aligned link + Relation-grounded link

Darling Mills Creek



"... part of the Parramatta River catchment ..."

Offline indexed facts
(h, r, t)

Relation-grounded link

Endpoint-aligned link

Parramatta River



"... main tributary of Sydney Harbour ..."

- Passage as retrieval state
- Endpoint-aligned link
- Relation-grounded link
- Source passage

Coarse-Grained Evidence Induction



Question anchors $A(Q)$

Darling Mills Creek

river

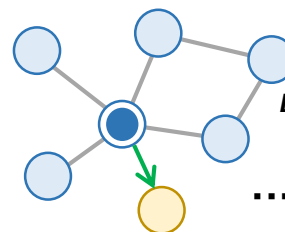
flow into

Dense entries
(seed set S_Q)

+

Subgraph
Induction

Query-local subgraph



Bounded-BFS

Candidate pool C_Q

- Darling Mills Creek
- Toongabbie Creek
- Murrumbidgee River
- Doubtful Creek
- Parramatta River
- ...
- 72 Parramatta River
- ...

Evidence Link grounded

Fine-Grained Evidence Retrieval



Mining Evidence needs $B(Q)$

s_1 : What river does Darling Mills Creek lead to?

s_2 : What does that river flow into?

Need support

"that river" $\approx$ Parramatta River

Evidence-conditioned
Reranking

Selected evidence set R_q

- Darling Mills Creek
- Murrumbidgee River
- Doubtful Creek
- Parramatta River
- Parramatta River
- ...



LLM

Sydney Harbour

Admitted by noisy-OR coverage

Reranking candidate pool for Evidence Complementarity

Step 1

Darling Mills Creek

Evidence link

Step 2

Parramatta River

Evidence link

Step 3 ✓

Sydney Harbour

distractor path

Toongabbie Creek

Evidence Lack



gold evidence transition

distractor

key discovery

Final answer